

Himanshu Pratap Rana

Software Engineer — Java / Spring Boot / Full-Stack Development

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SUMMARY

Software Engineer with hands-on experience in Java, Spring Boot, REST APIs, and backend development through internship and projects. Built applications using microservices, Kafka, RabbitMQ, MySQL, Docker, and application security practices. Experienced in AI/GenAI development with Python, LLMs, RAG, and AI agents. Strong foundation in DSA, OOP, DBMS, and distributed systems, focused on building scalable and production-ready software.

WORK EXPERIENCE

School Saavy

March 2025 – September 2025

Associate Engineer Intern

Gurugram, India

- Built a multi-tenant **school management SaaS** using **Java, Spring Boot, and PostgreSQL/MySQL**, supporting 5+ roles (Admin, Teacher, Student, Parent, Super Admin) with school-level data isolation.
- Designed and delivered **20+ RESTful APIs** for core modules including students, classes, assignments, and analytics, and engineered a configurable **assessment and grading system** with bulk result entry, improving average API response time and cutting manual administrative effort.
- Implemented **Role-Based Access Control (RBAC)** and a generic file management module with type validation, securing **100%** of protected endpoints across 3+ content types, and wrote unit tests with **JUnit and Mockito** to validate core business logic and catch regressions before deployment.
- Participated in **Agile/Scrum ceremonies**—sprint planning, stand-ups, and retrospectives—while containerizing services with **Docker** and managing **Git/GitHub** in a cross-functional team.

PROJECTS

Adaptive AI Developer Assistant | In Progress

| *Python, FastAPI, LLM, RAG, Agents, Evaluation*

- Built an AI-powered developer assistant using **Python and FastAPI** that analyzes developer queries, resumes, and technical documents to provide contextual recommendations.
- Implemented **RAG** with embeddings and vector search to retrieve relevant information before generating **LLM** responses.
- Developed tool-using AI agents and feedback/evaluation workflows to improve response quality and reduce unreliable outputs.
- Added persistent storage for conversations, feedback, and evaluation results, creating a foundation for continuous improvement.

Distributed E-Commerce Platform | In Progress

| *Java, Spring Boot, Microservices, MySQL, Kafka, RabbitMQ, Docker*

- Built a distributed e-commerce backend using **Java** and **Spring Boot** with independently deployable services for users, products, orders, payments, and inventory.
- Implemented **Kafka** for high-throughput, event-driven order and inventory workflows across services.
- Used **RabbitMQ** for asynchronous notification and email delivery, decoupled from the core order/inventory event stream.
- Secured APIs using Spring Security and JWT, and containerized services with Docker.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL

Core CS: DSA, OOP's, Operating Systems, DBMS, Computer Networks

Backend & Distributed Systems: Spring Boot, FastAPI, Microservices, REST APIs, Kafka, RabbitMQ, API Gateway, Event-Driven Architecture

AI/GenAI: LLMs, RAG, Embeddings, Vector Databases, AI Agents, Tool Calling, Prompt Engineering, LLM Integration, AI Evaluation

Database & Storage: MySQL, Vector Database

Deployment & DevOps: Docker, Docker Compose, Kubernetes, CI/CD

ACHIEVEMENTS

Solved **400+ Data Structures and Algorithms (DSA) problems** across LeetCode, CodeChef, and Codeforces. Profile: codolio.com/hhimanshu.

EDUCATION

Noida Institute of Engineering and Technology

Greater Noida, India

Bachelor of Technology (B.Tech), Computer Science and Engineering

2022 – 2026